

Detecting the Anomalous activity of a ship's engine

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1.Introduction

In this mini project, various anomaly detection methods are used to find the anomalous activity of a ship's engine. ML models were used to find the anomaly and to find the best ML method to detect the anomaly.

2.Problem Statement

In order to develop a machine learning-based anomaly detection system to identify abnormal patterns in ship operational data. The system should be capable of flagging anomalies in various data streams collected from ships, such as Engine rpm, Lubrication oil pressure, Fuel pressure, Coolant pressure, Lubrication oil temperature, Coolant temperature. By developing a robust anomaly detection system to protect a company's shipping fleet by evaluating engine functionality to work on downtime causing delayed deliveries can be improved. Consequently impacting the business revenue can be improved. By predicting timely maintenance, from evaluating ship's functionality the business aims to increase profit by, reducing safety risks for the crew, limiting fuel consumption, and increasing customer satisfaction through timely deliveries.

3.Description Analysis of Data

Necessary libraries(Pandas, numpy, seaborn and matplotlib.pyplot) were imported. CSV file was converted to a dataframe and few frames were displayed. Data shape was viewed and found to contain 19535 rows and 6 columns. Data was checked for missing values and found to be null. Data has been further checked for duplicate values and found to be none. Descriptive statistics have been done and the mean, standard deviation, minimum values, maximum value, 25th, 50th, 75th and 95th percentile were calculated. Histogram was created using seaborn. Histogram is shown in figure 1

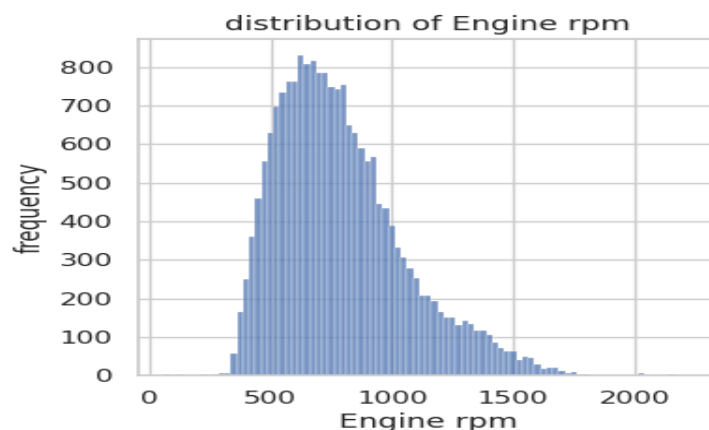


Figure 1: Histogram of Distribution of Engine rpm

4.Methods

Various methods were used to detect the anomaly in the data. IQR method, One class SVM and Isolation forest methods are used.

4a.IQR Method

For the IQR method, the data was sorted. Box plot using seaborn was created to visualise the data in each column. Box plot for engine rpm was shown in figure 2.

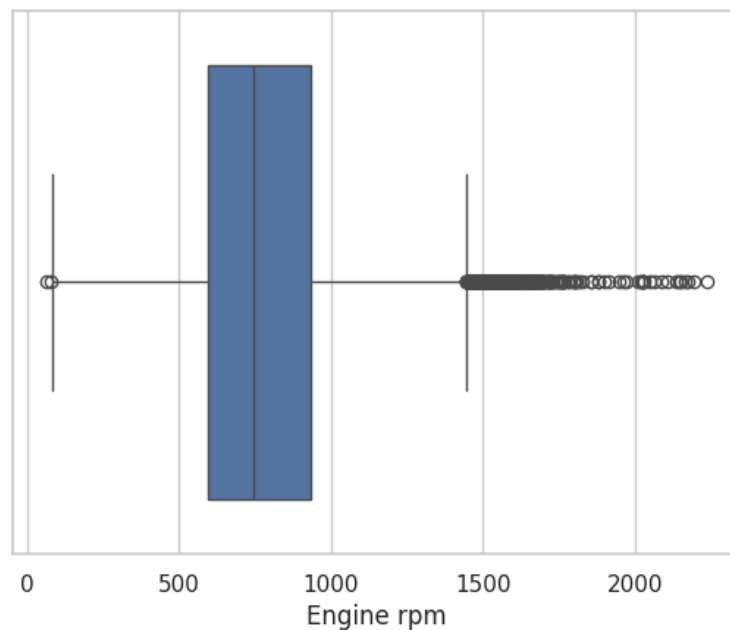


Figure 2:Box plot of Engine rpm

To Calculate IQR, 25th and 75th quantile were calculated. Using the quantiles, upper and lower limits were calculated. Data frames were created to from the values of the IQR method. The values outside the upper and lower limit were marked as anomalous. Prediction column was created for the sample which has anomalous value in more than one column, likewise more the three and four column values are simultaneously checked for anomalous activity.

Results

Percentage was calculated and it was found to be 2.16% for two features and 0.0536% for three features having anomalous activities. It's mentioned that the data has 1 to 5% anomalous values and IQR method findings were well within the range.

4b. One class Support Vector Model

An Unsupervised Method was performed on the data provided. Data was scaled using min max scaling method. Histogram was drawn to visualise the data before and after scaling using seaborn. Shown in figure 3.

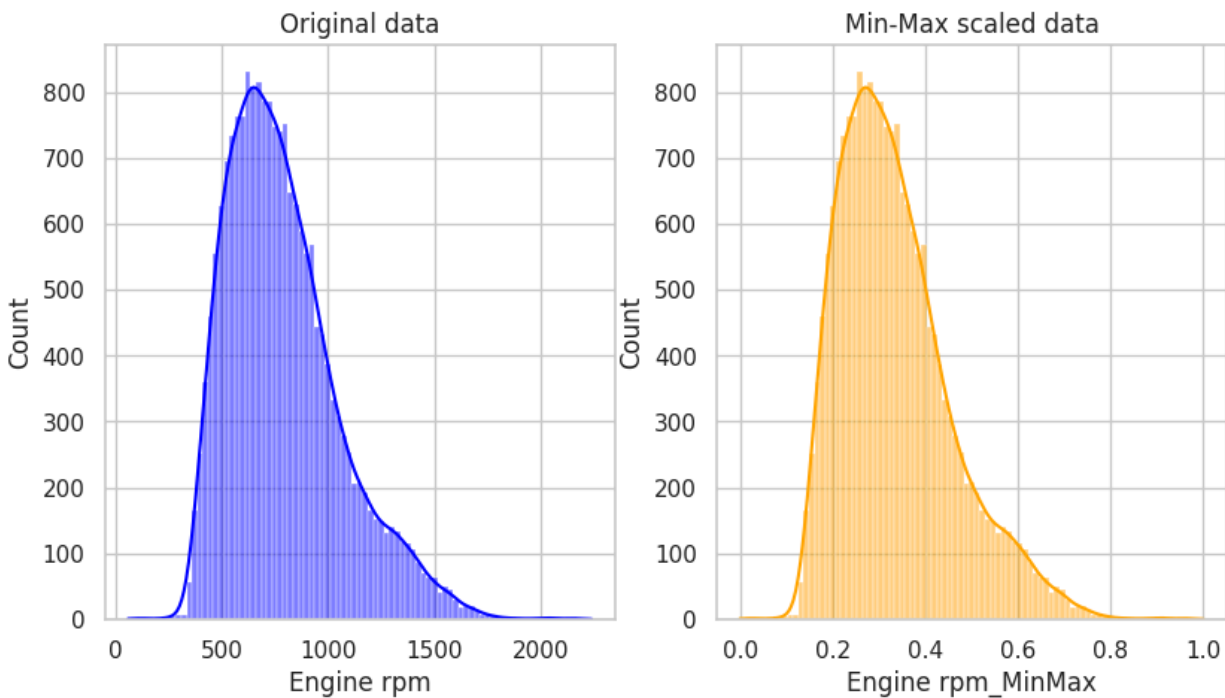


Figure 3: Showing histogram for Engine rpm and scaled Engine rpm

Scaled data has been used in the One class SVM model to predict the anomalies. Gamma values and Nu values were tuned to get the range of 1 to 5% anomalous data points.

Result

Decreased gamma value gives 975 rows of anomalous activities which is 4.99%. Similarly Gamma value is decreased, Nu value is also increased and decreased to predict the anomalies. It turned out to be 5.0, 3.99 and 3.0 respectively.

PCA was performed to visualise the data in a scatter plot. Six features were reduced to two components and scatter plot was created as shown in figure 4

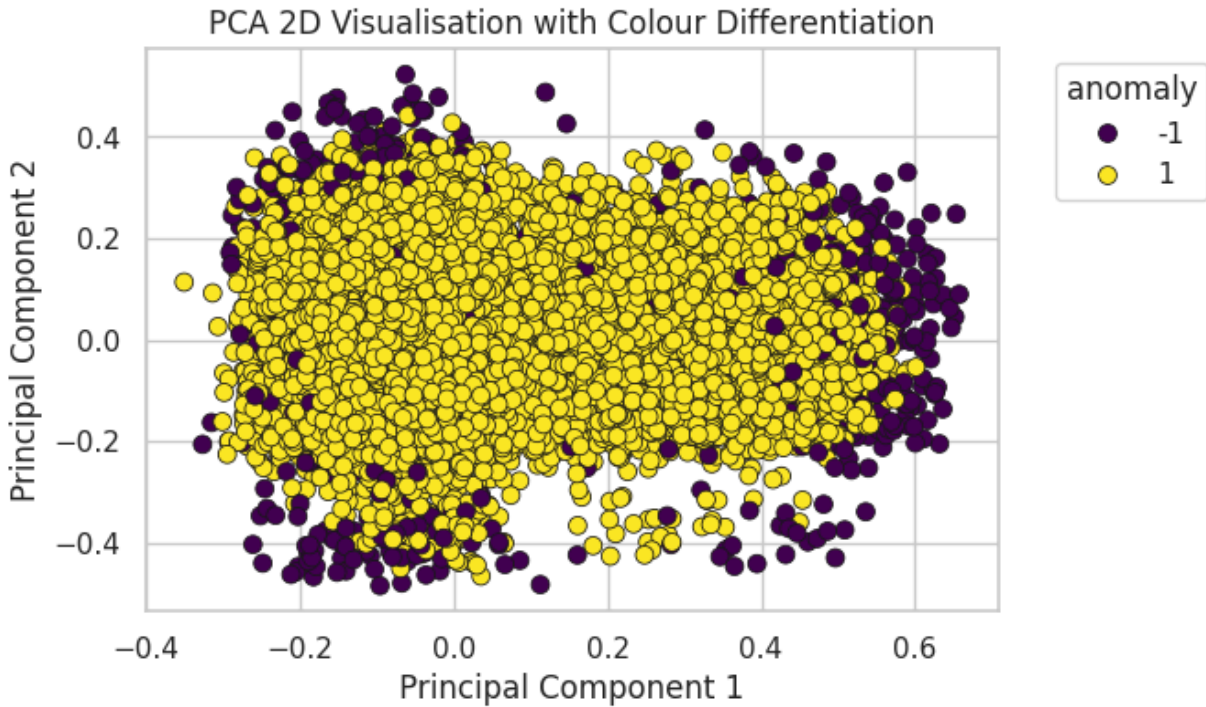


Figure 4: Scatter plot of PCA (One class SVM)

Visualise analysing the Scatter plot, it found to be lubricating oil pressure and lubricating pressure leads to anomalous values.

4c. Isolation Forest Method

Isolation forest method was performed and percentage of anomaly was calculated. It was found to be 5.00 and 3.00 with decreased contamination value. Data were scaled using min max scaling and PCA was performed to get two components for the Scatter plot for the Y value predicted from Isolation forest method.

Result

By analysing the scatterplot, it is found that the combination of lubrication oil pressure and fuel pressure results in anomalous values. The scatter plot from PCA is shown in Figure 5.

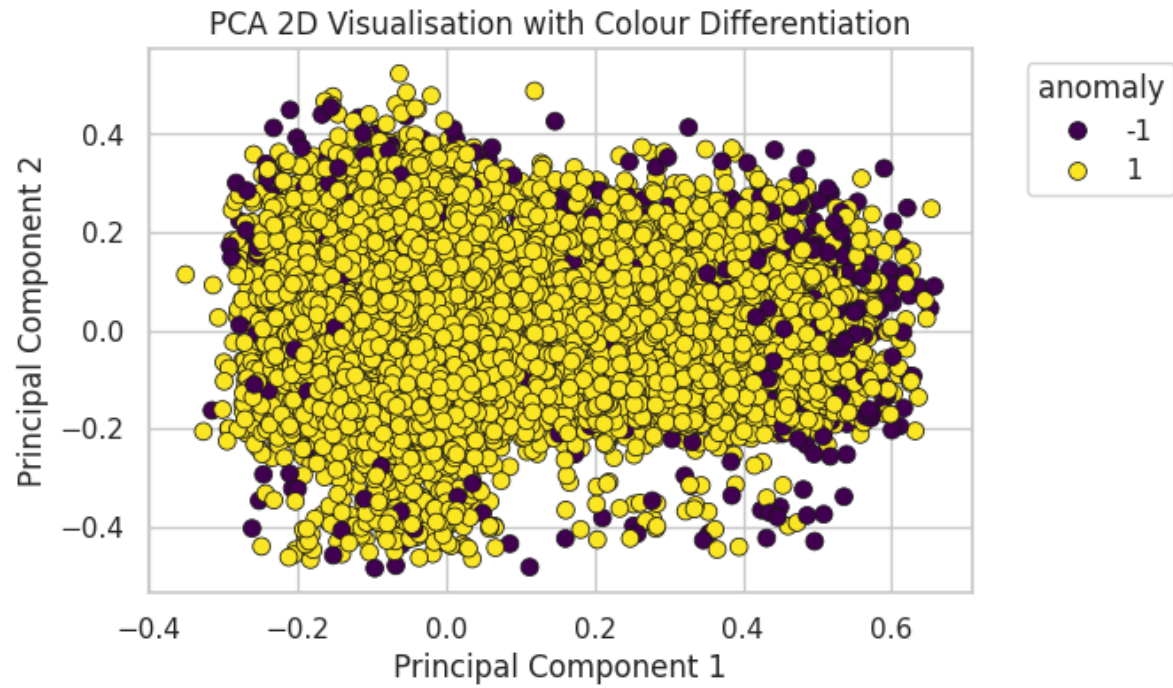


Figure 5: Scatter plot of PCA(Isolation Forest Method)

Conclusion

Hence the isolation forest method shows fewer anomalous values and is very robust in detecting anomalous values. Most of the values are clustered in the normal range.